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Review

Implications of Withaferin-A for triple-negative breast cancer chemoprevention

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ABSTRACT

Triple-negative breast cancer (TNBC) accounts for about 15 % of all breast cancer cases, and unlike other malignancies, it lacks definite prognostic markers. While improved survival responses have been documented with the ongoing therapeutic approaches, the development of tumor resistance mechanisms to these treatment options pose major challenges in the treatment of TNBC. Notably, naturally occurring medicinal compounds have been studied extensively for their anti-neoplastic activities in cancer models including breast cancer due to their safe and non-deleterious effects. Among various dietary compounds, Withaferin-A (WA), a phytochemical derived from an ayurvedic medicinal plant, *Withania somnifera* has been characterized to possess anti-inflammatory and anti-cancer properties. Importantly, multiple studies have shown that WA exhibits promising anti-tumoral activities against *in-vitro* and *in-vivo* experimental models of TNBC and that its combination has been documented to enhance chemotherapy efficacy. The current review highlights the mechanistic insights with recent updates including the pharmacokinetics parameters and implications of WA against breast cancer with major emphasis on TNBC.

1. Introduction

Breast cancer is one of the most predominant cancer types in women, and compared to other highly aggressive malignancies, a 0.3 % gradual increase in its prevalence has been noted annually [1,2]. On average, there is a 13 % risk of developing breast cancer in American women, sometime in their lifetime [1,3,4]. Notably, the prospective follow-up health study of nurse participants and population-based survey suggest that women with a family history or a first degree relative of breast cancer have an increased relative risk of developing breast cancer compared to women with no family history [5,6].

Histologically, breast cancer has been classified into multiple subtypes [7,8]. Based upon the site of origin, breast cancer is differentiated into carcinomas (i.e., epithelial origin), and sarcomas (i.e., stromal origin), respectively [8,9]. In most cases, cancer initiation begins in the ducts (i.e., ductal carcinoma), and glands (i.e., lobular carcinoma) of the breast, which make these carcinomas the predominant types [1,9]. However, sarcomas are a rare type of breast cancer [1,9]. Based upon the prognosis and the levels of gene expression, this cancer type has been classified into three major subtypes, namely, positive for estrogen and progesterone receptors (ER^+/PR^+) and human epidermal growth factor receptor 2 ($HER2^+$), which together constitute over two-thirds of all breast cancer cases, and TNBC accounts for about 15 % of all breast cancers, respectively [1,9–12].

In TNBC, the biomarkers, including ER, PR, and HER-2 gene amplification turn negative, which makes it a triple-negative and a highly aggressive malignancy [10–12]. Importantly, due to the heterogenicity, absence of the well-defined oncogenic drivers, and the ability to metastasize at distant organs to form secondary tumors, the diagnosis, and treatment of TNBC has been challenging. Of note, this cancer type has been documented to likely occur in women with mutations in the BRCA1 gene. Additionally, many signaling pathways such as EGFR, VEGF, mTOR are found to be overexpressed in TNBC [13,14].

Consistent with the other malignancies, the successful management of breast cancer types commonly includes its earlier detection at the benign stage, improved screening techniques, and advances in the

Abbreviations: WA, Withaferin-A; WLA/WLD, Withanolide A; WE, Withanone; SFN, Sulforaphane; TNBC, Triple-negative breast cancer; EMT, Epithelial to mesenchymal transition; ROS, Reactive oxygen species; OPA1, Optic atrophy protein 1; DRP1, Dynamin-related protein 1; CIS, Cisplatin; PI3K, Phosphoinositide 3-kinase; NF-kB, Nuclear factor-kappa B; BRAC1, Breast cancer susceptibility gene 1; HSF1, Heat shock factor 1; HDACs, Histone deacetylases; DNMTs, DNA methyltransferases; PK, Pharmacokinetics.

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treatment modalities such as surgical procedures like lumpectomy and mastectomy, radiotherapy, chemotherapy, immunotherapy as well as a combination of two or more therapeutic agents. Importantly, such treatment options have been shown to improve the overall survival (OS) rate of breast cancer patients compared to other malignancies [9,12,15]. Advances in all these led to a decrease in death rates in breast cancer patients [1,2,15]. However, since most of the clinically relevant chemotherapeutics are not targeted, patients experienced more adverse side effects, and cancer cells begin developing resistance to such therapies. These events have been implicated in reduced effectiveness or failure of anti-cancer drugs [9,12–15]. Thus, the development of novel therapeutics is necessary to overcome tumor resistance mechanisms and/or enhance the efficacy of ongoing therapeutic regimens without exerting deleterious effects.

2. Withaferin A

For ages, natural compounds obtained from a wide variety of plant products, have been a great source of drugs, used in folk medicine to treat several ailments including cancer [16–18]. Notably, such medicinal compounds have been considered for their promising therapeutic efficacy, against health disorder(s) with reduced side effects. Among several medicinal plants, *Withania somnifera* is native to Asia and has been used in Indian traditional ayurvedic medicine [16,19,20]. The roots and leaf extracts of this plant are known to possess many pharmacological activities like anti-angiogenic, anti-inflammatory, anti-cancer properties [16,20,21]. Of various phytochemicals, Withaferin A (i.e., WA), Withanolide A (i.e., WLA), and Withanone (i.e., WE) are the major and well-characterized compounds of the *Withania somnifera* plant.

In particular, WA, a steroidal lactone, has been shown to exhibit promising anti-cancer properties via its ability to target multiple signaling pathways, and thus, it interferes with many tumor cell processes including epithelial to mesenchymal transition (EMT), cell differentiation and proliferation, cell invasion, angiogenesis, and metastasis [16,19,21–23]. Importantly, multiple studies have shown that WA also has the potential to avert chemotherapeutic drug resistance [16,19,21–23]. Several cellular targets of WA have been identified including BRCA1, HSF1, HER2, Notch2, epigenetic enzymes, vimentin, tubulin, pro-apoptotic Bax, and anti-apoptotic Bcl-2 proteins [16,19,22,23]. The modulation (i.e., increase or decrease expression) of these signaling pathways have been shown to induce tumor cell death via mechanisms including the prevention of EMT and inhibition of angiogenesis [19,22]. This review explores the underlying chemopreventive mechanisms of WA intending to highlight various *in-vitro* and *in-vivo* models, as well as approaches utilizing WA to augment chemotherapy efficacy in the treatment of breast cancer with major emphasis on TNBC. Importantly, the significance of WA stemming from the *in-vivo* pharmacokinetic studies, the associated ongoing challenges, and its clinical relevance have also been explored.

3. WA and TNBC

3.1. In-vitro studies

Multiple studies have demonstrated that WA treatment can inhibit various cellular activities of breast cancer cells such as cell survival, proliferation, migration, and invasion via mechanisms involving the generation of ROS, regulation of oncogenic and tumor-suppressive signaling pathways leading to the induction of apoptosis [19,22–24]. The summary of *in vitro* studies is given in Table 1. Studies by Liu and colleagues have demonstrated that WA treatment inhibits the expression of NF-κB and mTOR pathways in a dose-dependent (i.e., 0, 6, 12, and 24 μM, assessed at 24 h' time point) in MDA-MB-231 and MCF-7 human breast cancer cell lines [19]. These changes resulted in a significant induction of apoptosis, which was found to be associated with increased expression of Bax and decreased expression of Bcl-2 proteins [19].

Given that mitochondria are an important source of ROS, Hahm and colleagues determined the mechanism of WA-induced apoptosis using MDA-MB-231 and MCF7 breast cancer cell lines [22]. The authors demonstrated that WA treatment caused a dose-dependent (i.e., 0, 1, 2, and 4 μM, assessed at 4 h' time point) increase in ROS production (& DNA fragmentation and apoptosis, evaluated with 2.5 μM dose) in a process blocked by the overexpression of Cu and Zn-superoxide dismutase [22]. Notably, normal human mammary epithelial cells (HMEC) were resistant to these WA-induced effects, indicating the specificity of WA towards the cancer cells. WA-induced increased apoptosis was mediated via the activation of Bax and Bak, and accompanied by a dose-dependent (i.e., 0, 2.5, and 5 μM, assessed at 4 h' time point) inhibition of oxidative phosphorylation (& complex III activity, measured with 5 μM dose at 6 h) of the mitochondrial electron transport chain. Importantly, MDA-MB-231 and MCF-7 cell lines expressing mitochondrial DNA-deficient Rho-0 variants were found to be resistant to WA-induced ROS generation, impaired mitochondrial membrane

Table 1
Summary of in-vitro studies.

Drug(s)	Cell lines used	Target(s)	Research findings	Refs.
WA	MDA-MB-231	Bax, Bcl-2, and NF-kB/mTOR	Increased Bax and decreased Bcl-2 levels resulted in the inhibition of cell viability and induction of apoptosis	[19]
	MCF-7, MDA-MB-231	Mitochondria	Increased ROS generation resulted in impaired mitochondrial membrane potential causing apoptosis via Bax and Bak activation	[22]
	MCF-7, MDA-MB-231, SUM159	Mitochondrial fusion and fission proteins including OPA1 and DRP1.	Alterations in mitochondrial dynamics resulted in decreased mitochondrial complex III activity, inhibition of mitochondrial fusion and fission abilities, decreased expression of fusion and fission regulating proteins and increased apoptosis	[24]
	MDA-MB-231 and MCF-7	Epigenetic HDAC enzymes and DNMT	Inhibition of these enzymes resulted in the inhibition of cell growth	[26]
	SUM159, MCF-7, SKBR-3, and MCF-10A	β-tubulin	Downregulation of β-tubulin inhibited the cell cycle progression and metastasis	[32]
	MDA-MB-231	PI3K/Akt, survivin, and HSP90	Inhibition of PI3K/AKT signaling resulted in cell cycle arrest, reduced cell proliferation and increased apoptosis	[35]
	MDA-MB-231 and BT20	BRCA1 and HSF1	Decreased expression and stability of these proteins resulted in increased apoptosis	[42]
	MCF-7 MDA-MB-231, MCF7-B2, SKBR3, and BT474	ERBB2 and ERBB3	Inhibition of ERBB3 coupling with ERBB2 resulted in the attenuation of ERBB2 overexpression, and increased WA sensitivity	[45]
	MCF-7, MDA-MB-231, T47D, and H1229	Vimentin	Increased phosphorylation and depolymerization of vimentin resulted in decreased cell migration /invasion and apoptosis induction	[48]
WA + cisplatin	MCF-7, MDA-MB-231, and SUM159	ATR/CHK1	Downregulation of ATR and CHK1 expression resulted in the inhibition of cell viability and improved cisplatin sensitivity	[52]

potential, and apoptosis induction compared with their respective wild-type cell lines [22].

In another report, the same group extended their findings to evaluate the effects of alterations in the mitochondrial dynamics such as mitochondrial fission and fusion in WA-induced apoptosis using SUM159, MCF7, and MDA-MB-231 breast cancer cell lines [24]. WA treatment (evaluated with 2 μM dose at 24 h' time point) resulted in the decreased assembly of mitochondrial complex III only in SUM159 and MCF7 but not in other cell lines [24]. Besides, WA treatment caused alterations in mitochondrial dynamics via inhibiting mitochondrial fusion and fission abilities, as well as mitochondrial volume in a dose- and time-dependent manner (i.e., 2 and 4 μM at 3, 6, and 12 h' time points) compared to vehicle-treated MDA-MB-231 or MCF-7 cells. These WA-induced effects were found to be mediated via decreased expression of mitochondrial fusion and fission regulating proteins including full-length optic atrophy protein 1 (OPA1) and dynamin-related protein 1 (DRP1), which were associated with increased apoptosis [24]. Overall, these findings provided an intriguing mechanism of WA-induced apoptosis involving the alterations of both complex III activity and mitochondrial dynamics in breast cancer cells.

Of various cellular alterations, epigenetic modifications cause changes in the gene expression that involve mechanisms other than the perturbations in DNA sequences [25–27]. Notably, histone deacetylases (HDACs) and DNA methyltransferases (DNMTs) are some of the epigenetic enzymes that overexpress in breast cancer cells [25–27]. A study led by Royston and colleagues determined the combinatorial effects of WA and another natural compound, sulforaphane (SFN) on HDAC1 and DNMTs enzymes using MCF-7 and MDA-MB-231 breast cancer cell lines [28]. The combination of physiologically low achievable doses of WA (i.e., 1.0 μM) and SFN (i.e., 5.0 μM) for 3 days led to synergistic inhibition of the cell viability in MCF-7, but not in MDA-MB-231 cells, yet significantly increased apoptosis was observed in both MCF-7 and MDA-MB-231 cell lines compared to WA and SFN alone treatments. Interestingly, no significant effects of WA, or SFN, or their combination were observed on the survival of a non-tumorigenic normal mammary epithelial MCF10A cell line. Also, this combination led to a significant decrease in the mRNA and protein expression of HDAC1 in both the cell lines, yet no changes were noticed in HDAC activity [28]. Meanwhile, synergistic inhibitions in *DNMT1* and *DNMT3A* mRNA expression but DNMT3A and DNMT3B protein expression were observed in MCF-7 but not in MDA-MB-231 cells, with no substantial changes were noted in DNMT activity by WA and SFN combination in both MCF-7 or MDA-MB-231 cell lines compared to WA or SFN alone treatments [28]. Importantly, the combination treatment resulted in increased mRNA and protein expression of Bax and decreased levels of Bcl-2 in both the cell lines compared to vehicle or WA and SFN alone treatments [28]. These findings indicated that WA and SFN combination at low concentrations can act as potential inhibitors of key epigenetic enzymes to effectively inhibit breast cancer cell growth.

Among various cellular targets, β-tubulin has gained considerable attention as an attractive chemotherapeutic target due to its essential role in cell cycle progression during mitosis and cellular movement [29–31]. To that end, Anthony and colleagues performed a comparative study with WA and its C6, C7-epoxy derivatives, WE and WLA to test whether these compounds target β-tubulin to regulate the growth of SUM159, MCF-7, and SK-BR-3 breast cancer cell lines and compared their effects with control MCF10A cells [32]. The authors demonstrated that WA treatment significantly inhibits the proliferation of MCF-7 and SUM159 cell lines in a dose (i.e., 1, 2, and 4 μM) and time (i.e., 24, 48, and 72 h)-dependent manner. Also, significantly increased G2/M cell cycle arrest was noted with 2 μM concentration of WA at 8 and 24 h' time points in all the cell lines and found to be associated with decreased protein expression of β-tubulin, caused by the disruption of microtubule spindle morphology [32]. Notably, these WA-induced effects were not observed by the treatments of epoxy derivatives, and also not seen in the MCF10A cell line [32]. Importantly, mass spectrometry analysis revealed that the molecular structure of WA is a key determinant in covalently binding with cysteine 303 residue of β-tubulin. Overall, these data indicated that β-tubulin is more sensitive to be targeted by WA in breast cancer cell lines compared to the non-tumorigenic MCF10A cells, and provided a novel mechanism of WA-mediated breast cancer chemoprevention.

As overactivation of the PI3K/AKT signaling contributes to chemotherapeutic drug resistance in breast cancer [33,34], Wang and colleagues did a comparative study with two Withanolides (i.e., 4-HW and WA) using MDA-MB-231 cell line [35]. While both 4-HW and WA decreased cell viability in a dose (i.e., 0.5, 1, 2, 5, and 10 μM) and time (i.e., 24 and 48 h)-dependent manner, these compounds exhibited different kinetics of growth suppression suggesting that these possess distinct underlying mechanisms [35]. Further studies demonstrated that 4-HW treatment induced a G1 phase (at the doses of 0.5, 1, and 2 μM at 24 h' time point), however, the same dosing regimen of WA induced a G2/M phase cell cycle arrest. Also, both these Withanolides caused apoptotic cell death, whereas WA treatment was also found to induce necrosis. The analysis of the cell survival proteins revealed that 4-HW treatment resulted in a dose-dependent (i.e., 0.2, 0.5, 1, 2, and 5 μM at 24 h' time point) inhibition of anti-apoptotic proteins, survivin, Bcl-2, Bcl-xl, and XIAP. In contrast, WA treatment significantly attenuated XIAP expression, exhibited less effects on Bcl-2 and Bcl-xl proteins, but inhibited survivin protein expression at higher (i.e., 2 and 5 μM) concentrations [35]. While 4-HW treatment upregulated cyclin-dependent kinase (CDK) inhibitors, p21 and p27 in a dose-dependent (mentioned above) manner, WA treatment slightly increased p21 but had no significant effect on p27 expression. Mechanistically, these 4-HW-induced effects were found to be mediated via dose-dependent (mentioned above) inhibition of AKT, ERK, and NF-kB phosphorylation (activation). In contrast, WA treatment only dose-dependently inhibited NF-kB activation, had little or no effect on ERK levels, induced AKT phosphorylation at lower (i.e., 0.2–2 μM) doses but decreased at higher (i.e., 5 μM) concentration, and found to be a more potent inhibitor of HSP90 [35]. Overall, these findings indicated that the differential mechanisms of 4-HW and WA should be exploited with distinct genetic characteristics of breast cancer cells, and that WA should be combined with P13 K/AKT inhibitors to achieve its improved responses for treating TNBC.

Genetic alterations in the breast cancer susceptibility gene 1 (i.e., BRCA1) and heat shock factor 1 (i.e., HSF1) are associated with an increased risk, or susceptibility to tumorigenesis in models including TNBC, indicating that agents targeting these genes offer promising treatment options for their prevention [36–41]. To that end, Zhang and colleagues determined if WA can target these proteins in MDA-MB-231 and BT20 TNBC cell lines [42]. The studies demonstrated that WA treatment resulted in increased apoptosis in a dose-dependent (i.e., 0.25, 0.5, 1, and 2.5 μM assessed at 24 h' time point) manner with increased sensitivity towards BT20 cells and that this effect was found to be mediated via dose-dependent decreased expression of HSF1 and BRAC1 proteins [42]. This mechanism was further confirmed by studies demonstrating that in the presence of cycloheximide (a protein synthesis inhibitor) or MG132 (a proteasome inhibitor), WA-induced (i.e., 2.5 μM dose at 6- or 24 -hs' time points) decreased expression of HSF1 and BRAC1 proteins were partially restored. These findings indicated that WA induces protein denaturation and proteasomal degradation dependent mechanism for depleting HSF1 and BRAC1 proteins, and suggest that HSF1 and BRCA1 could be used as potential therapeutic targets for WA against TNBC.

Notably, receptor tyrosine kinases represent one of the major cascades and have been considered as the potential targets in cancer prevention. Of various family members, receptor tyrosine-protein kinase, ERBB2 has been shown to couple with its dimerization partner ERBB3, and their increased expression pose a high risk for breast cancer induction [43,44]. In a study, Barnette and colleagues evaluated the effects of ERBB2/ERBB3 overexpression on WA sensitivity in a panel of breast cancer cell lines, namely MDA-MB-231, MCF-7, MCF7-B2,

SK-BR-3, and BT474, which differ in ERBB2 and ERBB3 expression [45]. The cell proliferation data revealed that ERBB2 overexpressing cell lines SK-BR-3 and BT474 exhibited higher sensitivity of dose-dependent (i.e., 0.5–5 μM assessed at 48 h' time point) inhibition by WA compared to other cell lines. This increased sensitivity of WA on SKBr3 and BT474 cell lines was found to be mediated via decreased phosphorylation of ERBB3 and PI3K/AKT, and that these effects were not seen in other cell lines like the MCF-7 and MDA-MB-231 [45]. This mechanism was further confirmed using isogenic MCF7-B2 cell model that exhibited 3–4 folds higher expression of ERBB2 compared to MCF-7 cells., and that this isogenic model showed increased WA sensitivity (i.e., 10 μM dose at 4- or 8 -hs' time points) compared to parental MCF-7 cell line [45]. Overall, these findings indicated that ERBB2-ERBB3 signaling contributes to enhancing WA sensitivity against breast cancer.

Vimentin is a type 3 intermediate filament and a classic EMT protein required for cell motility and migration and its overexpression has been correlated with tumor metastasis in breast cancer prevention [46,47]. Thaiparambil and colleagues evaluated the role of vimentin in WA-mediated anti-invasive/apoptotic effects against breast cancer [48]. The authors demonstrated that WA treatments to MDA-MB-231, MCF-7, T47D, and H1229 cell lines resulted in a dose-dependent (i.e., 9, 27, 81, 240, 500, 720 nM to 2 μM assessed at 24 h' time point) inhibition of cell migration and that at lower doses (≤500nM), WA exhibited potent anti-invasive but weak cytotoxic and apoptotic effects [48]. As cell migration requires vimentin assembly, confocal imaging microscopic data revealed that WA treatment at 250 nM dose caused perinuclear vimentin accumulation and subsequent rapid depolymerization, which were mediated by its increased phosphorylation at ser56 and disassembly compared to the untreated control cells [48]. These findings suggested that WA could be used with pharmacological inhibitors of vimentin to achieve synergistic anti-invasive and/or apoptotic effects.

Given the promising anticancer properties of WA against TNBC, its effect was also evaluated in combination with a chemotherapeutic agent. As ATR/CHK1 overexpression plays a key role in modulating DNA damage responses leading to cancer and has also been implicated in inducing resistance to cisplatin (CIS) chemotherapy [49–51]. Hahm and colleagues evaluated WA effects on ATR signaling, and cisplatin sensitivity using MDA-MB-231 and SUM159 TNBC cell lines [52]. The experimental studies revealed that WA treatment (with 2μM dose assessed at 8- and 16 -hs' time points) caused inhibition of cell viability via decreasing the expression of ATR and CHK1 proteins. Besides, WA treatment in combination with CIS for 24 h resulted in significant inhibition of cell viability in MDA-MB-231(WA at 0.5 and 1 μM & CIS at 30 and 60 μM doses) and SUM159 (WA at 0.5 and 1μM doses & CIS at 15 μM dose) compared to WA and CIS alone treatments [52]. Importantly, the autophosphorylation of ATR induced by CIS treatment was also abolished by WA. Overall, these findings suggested that WA exhibits its growth inhibitory effect by inhibiting ATR/CHK1 pathway and that similar mechanisms can be exploited to improve the efficacy of CIS therapy.

3.2. In-vivo studies

The above summarized *in-vitro* studies provided the rationale to test the anti-cancer potential of WA in *in- vivo* models of breast cancer including TNBC. Notably, studies by Liu and colleagues demonstrated that WA treatment suppresses the growth of MDA-MD-231 tumor xenografts via similar mechanisms as observed in cell culture-based experimental systems [19]. As most commercial anti-tumor agents specifically target primary tumors but not tumor metastases, another study evaluated WA effects for its anti-metastatic properties against breast cancer [48]. Using a metastatic model of 4T1 spontaneous mouse mammary carcinoma cells, the authors demonstrated that WA treatments (i.e., 0.1, 0.5, 1, 2, and 4 mg/kg doses given intraperitoneally at every alternate day for 30 days) resulted in a dose-dependent inhibition of metastatic lung nodule formation and its growth with minimum toxic effects to normal lung tissue [48]. Mechanistically, this anti-metastatic effect of WA was found to be mediated via increased vimentin phosphorylation at ser56 residue and its disassembly [48]. The findings of some *in-vitro* studies are presented along with the *in-vivo* studies to highlight their importance. The summary of these *in-vivo* studies is given in Table 2.

As dysregulation in the levels of cytokines has been implicated in cancer progression and/or metastasis [53,54], a study led by Khazal and colleagues determined the chemopreventive mechanisms of WA, and its effects on cytokine levels using the model of TNBC MDA-MB-231 cells [54]. Consistent with the *in-vitro* effects of decreased cell viability, cell cycle arrest, and apoptosis induction, WA treatment resulted in reduced growth of tumor xenografts and its metastatic ability when given orally at a concentration of 300 mg/kg/day for five days a week for 8 subsequent weeks [54]. Importantly, WA did not exert any cytotoxic effects in normal breast cells, indicating its specificity for TNBC. Tumor cytokine profiling revealed decreased levels of various cytokines including CCL2 [54]. To further define the mechanism, the authors found that decreased CCL2 expression was associated with reduced tumor growth and metastasis, indicating its potential in the prevention of breast cancer progression.

While oncogenes favor tumor development, tumor suppressor genes control cancer growth via regulating the activation of oncogenes, and among these, Notch2 has been shown to play critical roles as a tumor growth suppressor in breast cancer models [55–57]. Studies by Kim and colleagues evaluated the therapeutic effects of WA, as well as the vulnerability of Notch2 in altering tumor growth using *in-vivo* xenograft models of MDA-MB-231 and SUM159 TNBC cell lines [58]. The knockdown of Notch2 resulted in the augmentation of tumor growth and decreased apoptosis in both xenograft models and that these effects were found to be mediated by the upregulation of several cytokines [58]. This evidence verified the functional role of Notch2 in altering breast cancer growth. Also, following WA treatment, apoptosis induction was restored, and the levels of active Notch2 were found to be increased [58]. Moreover, Notch2 knockdown-mediated up and downregulation of cytokines such as IL-6, IL-8, IL-15, and CXCL-1 was also found to be reversed by WA treatment. These studies indicated that Notch2 knockdown can promote *in-vivo* TNBC growth and that WA can avert all these functions, demonstrating its potent anti-cancer property.

3.3. Pharmacokinetics studies

In addition to exploring the anti-tumoral and anti-metastatic activities and the underlying mechanisms of WA, several studies have also

Table 2
Summary of in-vivo studies.

Drug (s)	*In-vivo* models	Target(s)	Research findings	Refs.
WA	MDA-MB-231 xenograft	Bax, Bcl-2, and NFkB/ mTOR	Increased Bax and decreased Bcl-2 levels with NFkB/ mTOR signaling resulted in decreased tumor growth	[19]
	4T1 spontaneous murine metastatic model	Vimentin	Depolymerization of vimentin inhibited EMT and tumor metastasis	[48]
	MMTV neu mice	ATR	Downregulation of ATR overexpression resulted in decreased tumor growth	[52]
	MDA-MB-231 xenograft	Bax, Bcl2, and CCL2 cytokine	Increased ROS generation, apoptosis induction, and downregulation of CCL2 expression resulted in the attenuation of tumor growth.	[54]
	SUM159 and MDA-MB-231 xenografts	Notch2	Restoration of Notch2 function resulted in tumor growth suppression	[58]

evaluated its systemic bioavailability via pharmacokinetics (PK) studies. To determine the fate of WA in living organisms, Thaiparambil and colleagues administered WA at a single dose of 4 mg/kg intraperitoneally in female BALB/c mice, which resulted in the inhibition of metastatic lung nodule formation/growth of 4T1 mammary carcinomas, and analyzed various pharmacokinetic parameters [48]. The results revealed that the maximum (or peak) plasma concentration (i.e., C_{max}) of WA in the blood plasma was 1.8 μM with a half-life (i.e., $t_{½}$) of 1.36 h, and the area under the curve (i.e., $AUC_{0\text{-}t}$) value that represents the variation of a drug concentration in blood plasma as a function of time was 1.09 μM. hour. The clearance (CL) value represents the measurement of plasma volume from which a drug is completely removed per unit time, which was 2.52 mL/min. The volume of distribution (i.e., V_d) represents the apparent volume into which the drug would be distributed to provide the same concentration that it is observed in the blood plasma, which was 14.8 L/kg [48].

Another study led by Patil and colleagues tested the pharmacokinetic parameters of WA and its analog, Withanolide A (WLD) after oral administration of *Withania somnifera* root aqueous extract (WSE) at a dose of 1000 mg/kg to female albino mice [59]. The data obtained for WA revealed the following characteristics, C_{max} of 16.69 ± 4.02 ng/mL, $t_{½}$ of 59.92 ± 15.90 min, $AUC_{0\text{-}t}$ of 1572.27 ± 57.80 ng/mL min, and CL of 274.10 ± 9.10 mL/min/kg [59]. Moreover, the pharmacokinetic parameters of WLD were as follows: C_{max} of 26.59 ± 4.47 ng/mL, $t_{½}$ of 45.22 ± 9.95 min, $AUC_{0\text{-}t}$ of 2458.47 ± 212.72 ng/mL min, and CL of 191.10 ± 16.74 mL/min/kg [59]. These data demonstrated that the relative oral bioavailability of WLD was 1.44 times greater than WA, which could be attributed to their different physicochemical properties, or the PK interaction of their chemical constituents, and suggested careful evaluations of PK parameters of WSE or its constituents for the exploitation of their beneficial clinical applications.

Also, Dai and colleagues conducted PK studies to determine the oral bioavailability of WA in male Sprague-Dawley rats [60]. In this study, WA was given via intravenous injection at a dose of 5 mg/kg or an oral administration at a dose of 10 mg/kg to male Sprague-Dawley rats, and blood plasma samples were collected at various time points (0–24 hours) for the PK analysis. The findings revealed that the mean plasma concentration of WA decreases in a time-dependent manner and that it gets rapidly metabolized by liver microsomes with a half-life of approximately 5.6 min, which influences its oral bioavailability [60]. The perfusate concentration-time profiles of WA suggested that most of WA might be getting biotransformed or bound to the tissue following its pass-through from the intestinal wall and liver. The remaining WA which entered into the systemic circulation in the body was found to be 27.1 % within an hour of oral administration and that this was estimated to be consistent with the *in vivo* oral bioavailability of 32.4 ± 4.8 % [60]. Overall, these studies indicated that WA metabolism could be the main limiting factor in achieving good oral bioavailability *in vivo* and that this observed effect was also predicted to be similar in humans.

Although unrelated to breast cancer, WA effects have also been tested in human studies to assess its bioavailability and safety parameters. Pires and colleagues conducted a phase 1 clinical trial to determine the maximum tolerated dose (MTD) of a standardized root extract of *W. somnifera* containing 4.5 % of WA w/w in 13 advanced stage high-grade osteosarcoma patients [61]. The patients were divided into 4 cohorts (3 patients/cohort, except cohort 3) and the starting dose of *W. somnifera* extract used was 1600 mg (containing 72 mg of WA) which gradually escalated to 4800 mg/day (equivalent to 216 mg WA/day). The dose-escalation study was conducted with 4 dose-escalation cohorts corresponding to WA doses of 72, 108, 144, and 216 mg/day. Overall, the data suggest that WA formulation for up to 4800 mg/day (i.e., 216 mg WA/day) was well tolerated by patients and did not result in any dose-limiting toxicity [61]. While no grade 3 or grade 4 adverse events were documented, the common noticeable side effects include elevated liver enzymes and skin rash with other observed effects were fatigue, fever, edema, and diarrhea [61]. Notably, the PK analysis revealed no detectable levels of WA in the patients' samples, which indicates that WA can be rapidly absorbed, metabolized, and eliminated, and that could influence its oral bioavailability.

4. Ongoing challenges or limitations

Despite recent advances in therapeutic modalities, the development of tumor resistance mechanisms and treatment-related adverse events remained one of the major challenges in the treatment of breast cancer including TNBC [2,16]. This led to increased interest and exploration of phytochemicals possessing anti-cancer properties. However, one of the major obstacles remains in the characterization of such phytochemicals as well as the implementation of the high-throughput screening to scale-up the bioactive lead compounds [62–64]. Notably, the optimum dosing regimens and bioavailability of these natural agents need to be verified in preclinical models prior to their evaluation in human subjects. Overall, detailed pharmacokinetics and pharmacodynamics studies are needed to be thoroughly and precisely performed to optimize the biologically active and achievable doses of such phytochemicals including WA to evaluate their systemic effects before their exploration in human studies [53,54,63].

5. Future perspectives

The development of targeted therapy and immunotherapy approaches that target several cancer cell phenotypes including stem cells as well as immunoevasion mechanisms are being explored for the treatment and management of human malignancies including breast cancer [9,12]. While detailed mechanistic studies are underway to define the bystander effects of targeted therapies including the induction of tumor resistance mechanisms, the overall improved responses have been documented with reduced adverse effects in breast cancer patients [62,63]. Several combination approaches including with dietary agents are also being explored in preclinical settings with an overall goal of enhancing the efficacy of conventional and targeted therapeutic approaches. However, oral bioavailability limits the use of many bioactive phytochemicals for their beneficial application in the prevention and treatment of cancer. Therefore, deciphering novel strategies with natural compounds including WA to overcome this bioavailability issue, and to determine the pharmacological therapeutic doses applicable to a diverse setting of TNBC patients are some of the major concerns as well as future perspectives in the treatment of breast cancer [64–67].

6. Conclusion

As delayed diagnosis, severe adverse outcomes, and the development of tumor resistance mechanisms to ongoing therapies pose serious challenges in the treatment of breast cancer, dietary compounds represent promising approaches for cancer chemoprevention [64–67]. The leaves and roots extract of *Withania somnifera* are found to have a potent bioactive compound, WA which has been shown to possess several biological activities including anti-inflammatory, antioxidant, and anti-carcinogenic properties against various *in-vitro* and *in-vivo* models of breast cancer. Such WA-induced effects have been found to be mediated via its ability to target multiple cellular signaling pathways as well as cellular phenotypes. Importantly, the combination of WA has been found to enhance the anti-tumor activity as well as the sensitivity of anti-neoplastic drugs such as cisplatin to breast cancer cells. Taken together, the above-mentioned studies suggested a promising chemopreventive property of WA against breast cancer including TNBC.

Declaration of Competing Interest

The authors declare no conflict of interest.

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